

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2024-2025
M.Tech. in AIML
End-Semester Test
(EC-2 Regular Paper)

Course No. : AIMLCLZG525
Course Title : Natural Language Processing
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 30-03-2025_AN

No. of Pages	= 4
No. of Questions	= 7

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Question 1:

You are given 5 sentences, using these sentences you have to train a neural Network which will predict the missing words in following fill in the blank:

India won Champions trophy 3 times.

India's latest Champions trophy win was captioned by Rohit Sharma.

India played the Champions trophy final in 2025.

India defeated New Zealand in the Champions trophy final in 2025.

India is only country with 3 Champions trophy wins.

Indian cricket team ____ Champions Trophy in 2025.

By making suitable assumptions, design a neural net with 1 hidden layer(3 neurons) to find the missing word.

1. Draw this neural net, label the inputs/outputs clearly [1.5].
2. Show the dimensionality of each layer and also show the dimensionality of matrices used to transform the input to output across all layers [2.5].
3. Also, write equations used for each layer [1].

Note: Input features to be extracted based on a window of size-2 with bidirectional span.

Solution:

Step 1. find the vocabulary of words from these sentences [1]

Possible outputs: [2025, 3, Champions, India, New, Rohit, Sharma, Zealand, by, captioned, country, defeated, final, in, is, latest, only, played, the, times, trophy, was, win, wins, with, won]

Step 2. Show the input feature words into neural net

Inputs: [cricket, team, Champions, Trophy]

Step 3. Write following equations [1]

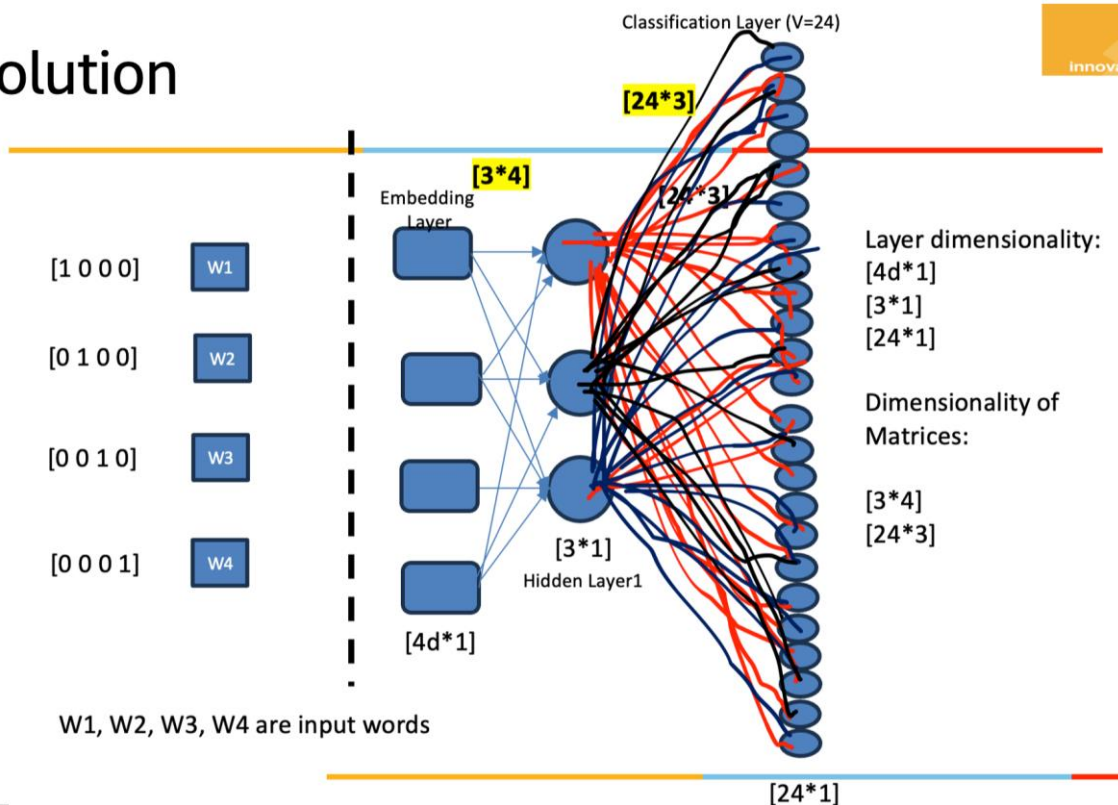
$$\mathbf{e} = [E_{w-2} \ E_{w-1} \ E_{w+1} \ E_{w+2}]$$

$$\mathbf{h} = \sigma(\mathbf{W}\mathbf{e} + \mathbf{b})$$

$$\mathbf{z} = \mathbf{U}\mathbf{h}$$

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{z})$$

Solution



Question 2 (3 + 2 = 5 Marks)

Part A: Part-of-Speech Tagging (3 Marks)

A simple part-of-speech tagger has the following transition and emission probabilities:

Transition Probabilities:

Current/Next	NN (Noun)	VB (Verb)
START	0.6	0.4
NN	0.3	0.7
VB	0.8	0.2

Emission Probabilities:

Tag/Word	"time"	"flies"
NN	0.7	0.4
VB	0.3	0.6

For the phrase "time flies", calculate the most likely POS tag sequence using the Viterbi algorithm. Show your work.

Part B: Hidden Markov Models (2 Marks)

A weather prediction system uses a Hidden Markov Model with two states: Sunny (S) and Rainy (R). The model parameters are:

Initial Probabilities: $P(S) = 0.7$, $P(R) = 0.3$

Transition Probabilities:

From/To	S	R
S	0.8	0.2
R	0.4	0.6

Emission Probabilities:

Observation/State	S	R
"Dry"	0.8	0.3
"Humid"	0.2	0.7

If we observe "Dry" on day 1, what is the probability of each weather state (Sunny or Rainy)

Solution (3 + 2 = 5 Marks)

Part A: Part-of-Speech Tagging (3 Marks)

For the phrase "time flies", using the Viterbi algorithm:

1. **Initialization (t=1, word "time"):**

- $V(NN, 1) = P(\text{START} \rightarrow NN) \times P(\text{time} | NN) = 0.6 \times 0.7 = 0.42$
- $V(VB, 1) = P(\text{START} \rightarrow VB) \times P(\text{time} | VB) = 0.4 \times 0.3 = 0.12$

Backpointers: both point to START

2. **For t=2, word "flies":**

- $V(NN, 2) = \max[V(NN, 1) \times P(NN \rightarrow NN) \times P(\text{flies} | NN), V(VB, 1) \times P(VB \rightarrow NN) \times P(\text{flies} | NN)] = \max[0.42 \times 0.3 \times 0.4, 0.12 \times 0.8 \times 0.4] = \max[0.0504, 0.0384] = 0.0504$ (from NN)
- $V(VB, 2) = \max[V(NN, 1) \times P(NN \rightarrow VB) \times P(\text{flies} | VB), V(VB, 1) \times P(VB \rightarrow VB) \times P(\text{flies} | VB)] = \max[0.42 \times 0.7 \times 0.6, 0.12 \times 0.2 \times 0.6] = \max[0.1764, 0.0144] = 0.1764$ (from NN)

3. **Termination:** Best final state = $\text{argmax}[V(NN, 2), V(VB, 2)] = \text{argmax}[0.0504, 0.1764] = VB$ Backtracking: $VB(2) \leftarrow NN(1)$

Therefore, the most likely POS tag sequence is: NN-VB.

Part B: Hidden Markov Models (2 Marks)

Using Bayes' rule to calculate the posterior probabilities:

1. For Sunny state given observation "Dry":

- $P(S | \text{Dry}) = [P(\text{Dry} | S) \times P(S)] / P(\text{Dry})$
- Numerator = $P(\text{Dry} | S) \times P(S) = 0.8 \times 0.7 = 0.56$

2. For Rainy state given observation "Dry":

- $P(R | \text{Dry}) = [P(\text{Dry} | R) \times P(R)] / P(\text{Dry})$
- Numerator = $P(\text{Dry} | R) \times P(R) = 0.3 \times 0.3 = 0.09$

3. Calculate $P(\text{Dry}) = P(\text{Dry} | S) \times P(S) + P(\text{Dry} | R) \times P(R) = 0.56 + 0.09 = 0.65$

4. Therefore:

- $P(S | \text{Dry}) = 0.56 / 0.65 = 0.862$ or 86.2%
- $P(R | \text{Dry}) = 0.09 / 0.65 = 0.138$ or 13.8%

Therefore, given the observation "Dry" on day 1, there is an 86.2% probability that the weather state is Sunny and a 13.8% probability that it is Rainy.

Question 3. [6 Marks]

Consider the sentence: "The apple fell from the tree"

Using the CFG given below 1) Create a Constituency Parse Table using the CKY algorithm 2) Draw a figure of the resulting parse tree.

- $S \rightarrow NP VP$
- $VP \rightarrow V Det N$
- $VP \rightarrow V P NP$
- $VP \rightarrow VP PP$
- $NP \rightarrow Det N$
- $PP \rightarrow P NP$
- $Det \rightarrow \text{"The"}$

- N -> "apple" | "tree"
- V -> "fell"
- P -> "from"

Solution to Question 3 [6 Marks]

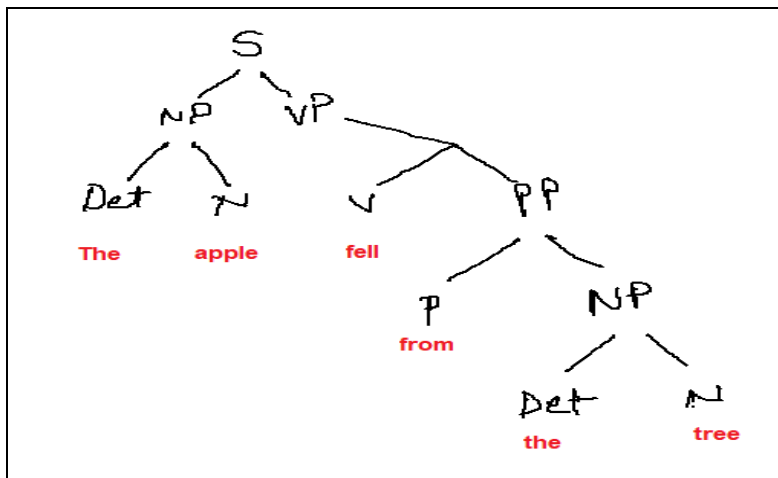
Evaluation guidelines:

- 2 marks for correctly converting the grammar into CNF form, as shown below

$S \rightarrow NP VP$
 $VP \rightarrow V NP$
 $VP \rightarrow V PP$
 $VP \rightarrow VP PP$
 $NP \rightarrow Det N$
 $PP \rightarrow P NP$
 $Det \rightarrow "The"$
 $N \rightarrow "apple" \mid "tree"$
 $V \rightarrow "fell"$
 $P \rightarrow "from"$

-
- 3 marks for correctly creating the CKY parse table shown below
- 1 mark for showing the parse tree figure

	1 The	2 <u>apple</u>	3 fell	4 from	5 the	6 <u>tree</u>
1	Det	NP				S
2		N				
3			V			VP
4				P		PP
5					Det	NP
6						N



Question 4. [6 Marks]

Consider the sentence: “The apple fell from the tree”

The following POS information is given about the words in this sentence:

The	- DET	(Determiner)
apple	- NOUN	(Noun)
fell	- VERB	(Verb)
from	- ADP	(Adposition/Preposition)
the	- DET	(Determiner)
tree	- NOUN	(Noun)

Now attempt the following questions:

Question 4a [1 Mark]

- Based on your understanding of “**Dependency Parse Tree**” graphically show, along with a brief explanation, the parse tree that is likely to result from the above sentence.

Question 4b [2 Marks]

- Explain in simple terms the transitions and the expected actions shown in the following Figure, which is related to the **ARC EAGER PARSING** algorithm.

$$\begin{aligned}
 \text{Left-Arc}(d) & \frac{([\dots, w_i]_S, [w_j, \dots]_B, A)}{([\dots]_S, [w_j, \dots]_B, A \cup \{(w_j, d, w_i)\})} \rightarrow \text{HEAD}(w_i) \\
 \text{Right-Arc}(d) & \frac{([\dots, w_i]_S, [w_j, \dots]_B, A)}{([\dots, w_i, w_j]_S, [\dots]_B, A \cup \{(w_i, d, w_j)\})} \\
 \text{Reduce} & \frac{([\dots, w_i]_S, B, A)}{([\dots]_S, B, A)} \rightarrow \text{HEAD}(w_i) \\
 \text{Shift} & \frac{([\dots]_S, [w_i, \dots]_B, A)}{([\dots, w_i]_S, [\dots]_B, A)}
 \end{aligned}$$

Question 4c [3 Marks]

- Using the Table format given below, show ONLY the first 6 transitions likely to be taken by the Arc Eager Parsing algorithm to parse the above sentence in order to create the dependency parse tree.

Step No	Transition	Stack	Buffer	Arcs

Solution to Q4

Solution to 4a [1 Mark]

<pre> graph TD ROOT --> fell fell --> apple fell --> from apple --> The from --> tree from --> the </pre> Explanation: • 'fell' is the main verb in the statement • 'apple' is the subject • 'from the tree' is the prepositional phrase: hence here is a link from 'fell' to 'from' and further on to 'tree'	Evaluation guidelines: • If ROOT is not shown, DO NOT cut marks • If 'tree' is shown with a direct link to 'fell' instead of 'from', DO NOT deduct marks • So long as explanation identifies 'fell' as the verb and 'apple' as the subject, give full marks.
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Solution to 4b [2 Marks – 0.5 marks for each correct transition listed and described approximately as follows]

- **LEFTARC:** Assert a head-dependent relation between the word at the front of the input buffer and the word at the top of the stack; pop the stack.
- **RIGHTARC:** Assert a head-dependent relation between the word on the top of the stack and the word at the front of the input buffer; shift the word at the front of the input buffer to the stack.
- **SHIFT:** Remove the word from the front of the input buffer and push it onto the stack.
- **REDUCE:** Pop the stack.

Alternate interpretation

Transition	Description
Left-Arc (X -> Y)	If the top of the stack 'Y' depends on the element 'X' below it in the Stack. X gets removed from the Stack.
Right-Arc (X->Y)	If the second element 'X' in the Stack depends on the top of the Stack 'Y'
Shift	Move a word from the Buffer to the Stack
Reduce	Remove the top of the Stack if it has already been assigned a head

Solution to 4c [3 Marks – 0.5 marks for each correct transition captured, the first 6]

- Either of the following two solutions, or equivalent is Ok.
- It is also Ok if the arcs are shown as follows: [apple->The] instead of [apple,det,The]

Solution-1

Step	Transition	Stack	Buffer	Arcs
1	Shift	Root, The	apple fell from the tree	
2	Left-Arc	Root	apple fell from the tree	[apple, det,The]

3	Shift	Root, apple	fell from the tree	[apple, det, The]
4	Right-Arc	Root, apple, fell	from the tree	[apple, det, The] [fell, subj, apple]
5	Shift	Root, apple, fell, from	the tree	[apple, det, The] [fell, subj, apple]
6	Shift	Root, apple, fell, from, the	tree	[apple, det, The] [fell, subj, apple]

Solution-2

Step	Action	Stack	Buffer	Arcs
1	Shift	The	apple fell from the tree	-
2	Shift	The apple	fell from the tree	-
3	Left-Arc (det)	apple	fell from the tree	(apple → The)
4	Shift	apple fell	from the tree	(apple → The)
5	Right-Arc (subj)	fell	from the tree	(apple → The), (fell → apple)
6	Shift	fell from	the tree	(apple → The), (fell → apple)

Question 5 [3+3=6 marks]

- a) Show how you would apply simple Lesk algorithm for lexical word disambiguation **[3 marks]**

The curtains shut out light from the **sun**

noun

Sense 1. the natural agent that [stimulates](#) sight and makes things visible. Ex: "the light of the sun"

Sense 2. an expression in someone's eyes indicating a particular emotion or mood. Ex: "a shrewd light entered his eyes"

verb

Sense 3. provide with light or lighting; [illuminate](#). Ex: "the room was lit by a number of small lamps"

Sense 4. make (something) start burning; ignite. Ex: "Allen gathered sticks and lit a fire"

adjective

Sense 5. having a considerable or sufficient amount of natural light; not dark. Ex: "the bedrooms are light and airy"

Sense 6. (of a colour) pale. Ex: "her eyes were light blue"

- b) Develop an ontology with following concepts [3 marks]:

Movie, Actress, Actor and Director

Also mention relations

- Movie1 directedBy director
- Movie1 has Actress
- Movie1 has Actor
- Actor acted In Movie2
- Actress acted in Movie3
- Director directs Movie3

Also mention disjoint relations if any

Q6 (3 + 3 = 6 marks)

[a] As per WordNet, the word "run" has 16 senses as Noun and 41 senses as Verb. Below are a selected few, with simplified contextual embeddings:

run_N^1	3. (2) {07358237} <noun.event> footrace#1, foot race#1, run2#3 -- (a race run on foot; "she broke the record for the half-mile run")	[-1, 2, -3]
run_V^1	4. (20) {02420190} <verb.social> operate#1, run2#4 -- (direct or control; projects, businesses, etc.; "She is running a relief operation in the Sudan")	[-1, -2, 3]

In a specific target sentence, BERT computed the contextual embedding of the word “run” to be [1, -2, 3].

Calculate the cosine similarities of the target with respect to each of the above senses and rank them in order of increasing cosine distance. (3 marks)

[b] Consider the sentence: “Hello World”

Let the arbitrary 2-dimensional vectors for these words be:

“Hello”: [0,1]

“World”: [1,0]

For simplicity assume Identity Matrix as the Projection Matrix, yielding:

$Q_Hello = K_Hello = V_Hello = [0,1]$

$Q_World = K_World = V_World = [1,0]$

Compute the updated representation of the 2 words “Hello” and “World” after applying self-attention.

Assume: $\text{Exp}(0) = 1$ & $\text{Exp}(1) = 2.72$

(All computations up to second decimal place only)

(3 marks)

Solution:

[a]

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

Cosine Distance = $1 - \text{Cosine Similarity}$

1. Magnitude of run_N^1

$$= \text{Sqrt}(\text{sqr}(-1) + \text{sqr}(2) + \text{sqr}(-3)) = \text{Sqrt}(14)$$

Magnitude of run_V^1

$$= \text{Sqrt}(\text{sqr}(-1) + \text{sqr}(-2) + \text{sqr}(3)) = \text{Sqrt}(14)$$

Magnitude of run_{target}

$$= \text{Sqrt}(\text{sqr}(1) + \text{sqr}(-2) + \text{sqr}(3)) = \text{Sqrt}(14)$$

2. Cosine Similarity of run_N^1 and run_{target}

$$\text{Dot Product} = (1 \times -1) + (-2 \times 2) + (3 \times -3) = -1 - 4 - 9 = -14$$

$$\text{Cosine Similarity} = -14 / (\text{sqrt}(14) * \text{sqrt}(14)) = -1$$

$$\text{Cosine Distance} = 1 - (-1) = 2$$

3. Cosine Similarity of run_V^1 and run_{target}

$$\text{Dot Product} = (1 \times -1) + (-2 \times -2) + (3 \times 3) = -1 + 4 + 9 = 12$$

$$\text{Cosine Similarity} = 12 / (\text{sqrt}(14) * \text{sqrt}(14)) = 0.857$$

$$\text{Cosine Distance} = 1 - (0.857) = 0.143$$

Rank in increasing order of cosine distance

$$run_V^1 < run_N^1$$

[b]

$$\text{Score}(\text{Hello}, \text{Hello}) = Q_Hello * K_Hello^T = [0,1] \cdot [0,1]^T = [0,1] \cdot [1,0] = 0$$

$$\text{Score}(\text{Hello}, \text{World}) = Q_Hello * K_World^T = [0,1] \cdot [1,0]^T = [0,1] \cdot [0,1] = 0 * 0 + 1 * 1 = 1$$

$$\text{Score}(\text{World}, \text{Hello}) = Q_World * K_Hello^T = [1,0] \cdot [0,1]^T = [1,0] \cdot [1,0] = 1$$

$$\text{Score}(\text{World}, \text{World}) = Q_World * K_World^T = [1,0] \cdot [1,0]^T = [1,0] \cdot [0,1] = 0$$

So, the Attention Score Matrix is:

[0 1

1 0]

Applying Softmax to obtain Attention Weights:

For the 1st row: $[\exp(0)/(\exp(0)+\exp(1)), \exp(1)/(\exp(0)+\exp(1))] = [1/3.72, 2.72/3.72] = [0.27, 0.73]$

Similarly, for the 2nd row, we have: [0.73, 0.27]

So the Attention Weight Matrix: [0.27 0.73

0.73 0.27]

Compute Weighted Sum of Value Vectors:

[0.27 0.73

[0 1

0.73 0.27]

X

1 0]

= [0.73 0.27

0.27 0.73]

So, the new representation

Question 7. Text Summarization [1+2.5+1+1.5 = 6 Marks]

For the generic text summarization, given below 5 documents in the collection as per the chronological order of mentions, answer the following questions sequentially.

Here are the documents:

Document 1 (d1): "Anna loves the ocean."

Document 2 (d2): "Anna visited the beach yesterday."

Document 3 (d3): "The waves at the beach were big and strong."

Document 4 (d4): "Tom enjoys surfing."

Document 5 (d5): "Anna and Tom spent the day together at the beach."

a) Preprocess the training document to:

a. Remove stop words { the, and, at, is, a, were}

b. Remove punctuations {'.', ',', '}'

c. Lemmatize all words

Note: Ignore word cases but retain all the words.

b) Compute the saliency score of every sentence using the sum of weights (normalized by size of pre-processed sentence) of all the words in the above preprocessed sentences. Use only the new simplified formula to compute word's weightage.

Weight (word W_i) = $TF(i,j) * N/DF(i)$

where,

N = Number of document given in the collection

TF(i,j) = Frequency of word i in document j

DF(i) = Number of documents containing word i

c) Extract summary in reverse chronological order with the top 2 most informative documents from part b)'s result.

d) Use ROUGE-1 to evaluate the results of the above system summary obtained in part b) w.r.t the below reference summary.

"Anna and Tom enjoy the beach together."

Solution:

Step 1: Preprocess the Documents

Preprocess each document by removing stop words, punctuations, and lemmatizing the words. Ignore word cases but retain all words.

Preprocessed Documents:

- Document 1 (d1): "anna love ocean"
- Document 2 (d2): "anna visit beach yesterday"
- Document 3 (d3): "waves beach big strong"
- Document 4 (d4): "tom enjoy surfing"
- Document 5 (d5): "anna tom spend day beach"

Step 2: Compute Word Frequencies and Document Frequencies

Calculate $TF(i,j)$ and $DF(i)$ for each word.

$TF(i,j)$ and $DF(i)$ Calculation:

- $TF(i,j)$: Frequency of word i in document j .
- $DF(i)$: Number of documents containing word i .

Word	TF in d1	TF in d2	TF in d3	TF in d4	TF in d5	DF(i)
anna	1	1	0	0	1	3
love	1	0	0	0	0	1
ocean	1	0	0	0	0	1
visit	0	1	0	0	0	1
beach	0	1	1	0	1	3
yesterday	0	1	0	0	0	1
waves	0	0	1	0	0	1
big	0	0	1	0	0	1
strong	0	0	1	0	0	1
tom	0	0	0	1	1	2
enjoy	0	0	0	1	0	1
surfing	0	0	0	1	0	1
spend	0	0	0	0	1	1
day	0	0	0	0	1	1

Step 3: Calculate Weight for Each Word

Use the formula: $Weight(word\ W_i) = TF(i,j) * N/DF(i)$, where $N = 5$.

Weight Calculation:

- Weight for each word in each document:
 - d1: anna = $15/3 = 5/3$, love = $15/1 = 5$, ocean = $1*5/1 = 5$
 - d2: anna = $15/3 = 5/3$, visit = $15/1 = 5$, beach = $15/3 = 5/3$, yesterday = $15/1 = 5$
 - d3: waves = $15/1 = 5$, beach = $15/3 = 5/3$, big = $15/1 = 5$, strong = $15/1 = 5$
 - d4: tom = $15/2 = 5/2$, enjoy = $15/1 = 5$, surfing = $1*5/1 = 5$
 - d5: anna = $15/3 = 5/3$, tom = $15/2 = 5/2$, spend = $15/1 = 5$, day = $15/1 = 5$, beach = $1*5/3 = 5/3$

Step 4: Calculate Saliency Score for Each Document

Sum the weights of all words in each document and normalize by the number of words.

Saliency Score Calculation:

- $d1: (5/3 + 5 + 5) / 3 = (5/3 + 10) / 3 = (5 + 30) / 9 = 35/9 \approx 3.89$
- $d2: (5/3 + 5 + 5/3 + 5) / 4 = (10/3 + 10) / 4 = (10 + 30) / 12 = 40/12 \approx 3.33$
- $d3: (5 + 5/3 + 5 + 5) / 4 = (20/3 + 10) / 4 = (20 + 30) / 12 = 50/12 \approx 4.17$
- $d4: (5/2 + 5 + 5) / 3 = (5/2 + 10) / 3 = (5 + 20) / 6 = 25/6 \approx 4.17$
- $d5: (5/3 + 5/2 + 5 + 5 + 5/3) / 5 = (10/3 + 5/2 + 10) / 5 = (20 + 15 + 60) / 30 = 95/30 \approx 3.17$

Step 5: Extract Summary

Select the top 2 most informative documents based on saliency scores in reverse chronological order.

Summary:

- Document 5 (d5): "Anna and Tom spent the day together at the beach."
- Document 4 (d4): "Tom enjoys surfing."

Step 6: Evaluate Summary Using ROUGE-1

Compare the generated summary with the reference summary: "Anna and Tom enjoy the beach together."

ROUGE-1 Calculation:

ROUGE-1 measures the overlap of unigrams between the system summary and the reference summary.

- System Summary: "anna tom spend day beach tom enjoy surfing"
- Reference Summary: "anna tom enjoy beach together"

Overlapping Unigrams:

- "anna", "tom", "beach"

ROUGE-1 Score:

$ROUGE-1 = \frac{\text{Number of overlapping unigrams}}{\text{Total number of unigrams in reference summary}}$

$ROUGE-1 = \frac{3}{5} = 0.6$

$ROUGE-1 = 0.6$

The final answer is:

- Preprocessed Documents: d1: "anna love ocean", d2: "anna visit beach yesterday", d3: "waves beach big strong", d4: "tom enjoy surfing", d5: "anna tom spend day beach"
- Saliency Scores: d1 ≈ 3.89 , d2 ≈ 3.33 , d3 ≈ 4.17 , d4 ≈ 4.17 , d5 ≈ 3.17
- Summary: d5: "Anna and Tom spent the day together at the beach.", d4: "Tom enjoys surfing."
- ROUGE-1 Score: 0.6